FULLY SYNTHETIC · INVENTED DATA · NO ECDLP SOLVES

An isogeny-volcano comparison experiment

Thirty, forty and fifty-bit field-size labels. Prime-field, ordinary binary and Koblitz-origin models. Ten symbolic targets at every vertex.

This is an executable simulation for planning a later experiment. Every cost, success probability, relation count, curve variation and comparison in the result tables is generated from declared assumptions. No curve coefficients, field representation, point coordinates, isogeny kernels, relation matrices or recovered logarithms were constructed.

The requested scope

Condition	Implementation
Sizes	$\log_2(q)$ labels 30, 40, 50; nominal prime r just below 2^(b−2), matched across families
Deep graph	Five strata, depth 4; vertex counts 1 / 4 / 12 / 36 / 108; 161 abstract vertices per condition
Horizontal connections	Three illustrative degree labels: 11, 23, 29. 483 overlay multiedges plus 160 vertical edges
Targets	10 holdout target labels per vertex and method; 3 separate pilot targets for destination selection
Methods	Rho; two-summand index calculus with a scan; three-summand index calculus with a pair table
Controls	Null; oracle scan cost ×0.55 per level; oracle scan cost ×1.7 per level; native weights 0 / 0.1 / 1
Accounting	Cold single-target cost; reusable precomputation shared across exactly 10 targets

Graph coverage means the complete specified abstract component, repeated for each family and size. It does not mean all elliptic curves or all isogeny degrees over any field.

What the simulation illustrates

The generated results can show how subgroup size, an imposed oracle trend, Frobenius discounts, memory and destination discovery costs interact. A size trend here reflects the equations in config.json. It provides no empirical evidence that real curves have the same trend.

Ten-target summaries at the symbolic surface

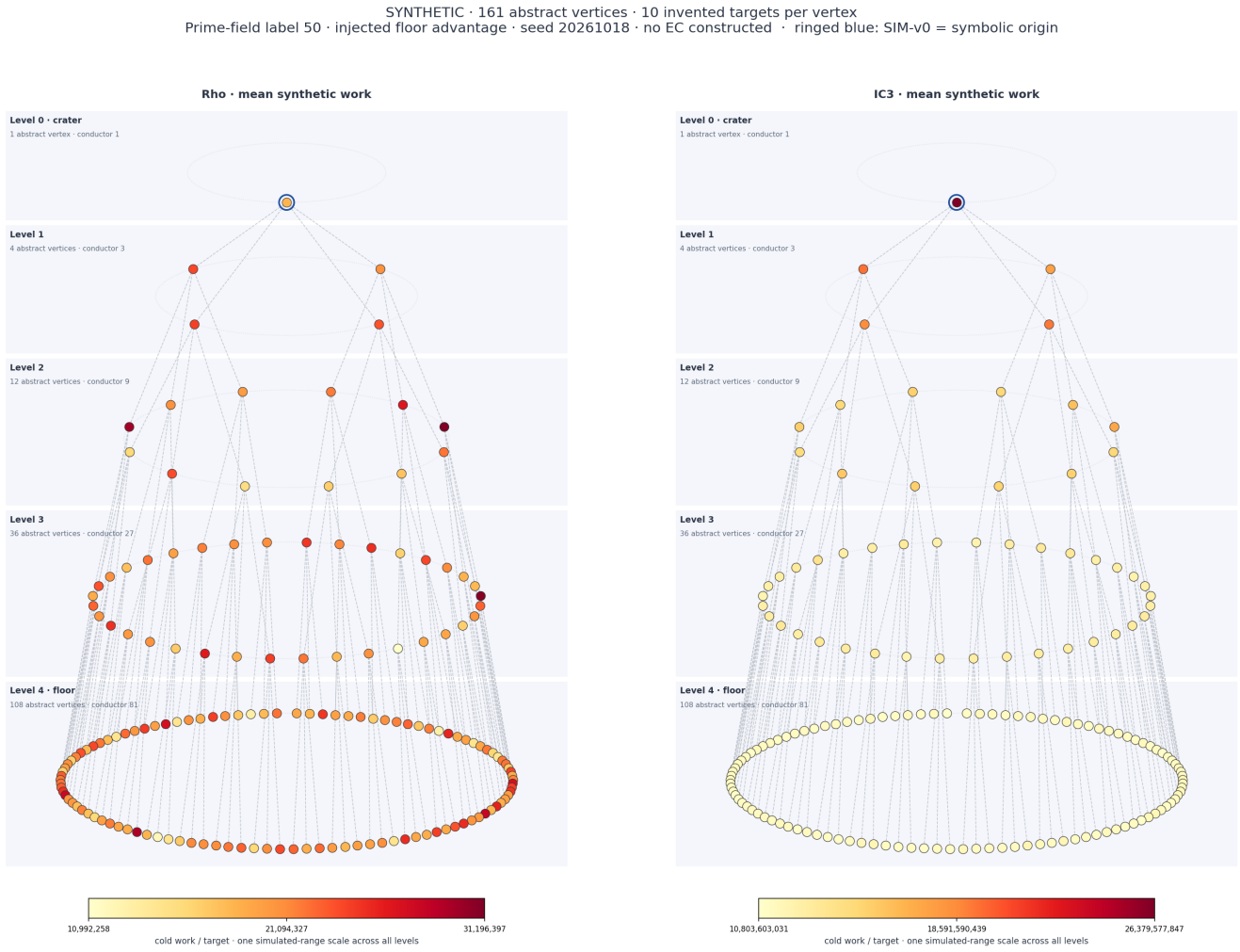
Null scenario; cold accounting; native-operation weight 0.1. All entries below are invented synthetic work units. The ratio is mean method cost divided by mean matched surface-rho cost; a larger value is more expensive.

Family	Bits	Method	Min	Mean	Max	Ratio / rho
Prime field	30	Rho	3.34e+03	1.87e+04	4.3e+04	1
Prime field	30	IC2 scan	3.38e+08	3.62e+08	3.94e+08	1.94e+04
Prime field	30	IC3 pair table	2.51e+06	2.65e+06	2.89e+06	142
Prime field	40	Rho	2.73e+05	4.95e+05	9.49e+05	1
Prime field	40	IC2 scan	3.46e+11	3.56e+11	3.72e+11	7.19e+05
Prime field	40	IC3 pair table	2.56e+08	2.6e+08	2.65e+08	524
Prime field	50	Rho	9.99e+06	1.83e+07	2.58e+07	1
Prime field	50	IC2 scan	3.62e+14	3.65e+14	3.69e+14	1.99e+07
Prime field	50	IC3 pair table	2.62e+10	2.64e+10	2.65e+10	1.44e+03
Ordinary binary	30	Rho	5.48e+03	1.87e+04	4.26e+04	1
Ordinary binary	30	IC2 scan	3.42e+08	3.78e+08	4.16e+08	2.02e+04
Ordinary binary	30	IC3 pair table	2.55e+06	2.71e+06	2.85e+06	145
Ordinary binary	40	Rho	2.64e+05	6.09e+05	8.86e+05	1
Ordinary binary	40	IC2 scan	3.72e+11	3.81e+11	3.91e+11	6.25e+05
Ordinary binary	40	IC3 pair table	2.65e+08	2.72e+08	2.75e+08	446
Ordinary binary	50	Rho	9.87e+06	1.82e+07	2.95e+07	1
Ordinary binary	50	IC2 scan	3.81e+14	3.87e+14	3.91e+14	2.13e+07
Ordinary binary	50	IC3 pair table	2.73e+10	2.74e+10	2.75e+10	1.51e+03
Koblitz origin	30	Rho	5.21e+03	8.3e+03	1.21e+04	1
Koblitz origin	30	IC2 scan	1.05e+07	1.62e+07	1.94e+07	1.95e+03
Koblitz origin	30	IC3 pair table	3.44e+05	3.62e+05	4.04e+05	43.6
Koblitz origin	40	Rho	2.12e+04	1.38e+05	2.71e+05	1
Koblitz origin	40	IC2 scan	7.76e+09	8.81e+09	1.14e+10	6.37e+04
Koblitz origin	40	IC3 pair table	3.14e+07	3.2e+07	3.24e+07	231
Koblitz origin	50	Rho	9.77e+05	4.33e+06	7.97e+06	1
Koblitz origin	50	IC2 scan	6.33e+12	6.7e+12	7.03e+12	1.55e+06
Koblitz origin	50	IC3 pair table	3.09e+09	3.11e+09	3.13e+09	717

Full precision, medians and standard deviations are retained in strategy_summary.csv. curve_summary.csv has min/mean/max for each of the 161 vertices in every condition. level_summary.csv first averages vertices within each target, then summarizes the ten target-level means; this is a different estimand from per-curve target variability.

GAE here means an invented group-operation count plus weight × an invented native-operation count. The weights are an accounting sensitivity band, not a hardware calibration. Prime versus binary timings are not inferred from these uncalibrated counts.

The deep graph and horizontal overlays



The heat map follows the repository’s standard volcano layout. All displayed links are dashed because none is a verified isogeny. The interactive explorer contains the full labeled multigraph, including same-stratum overlay edges and surface self-loops.

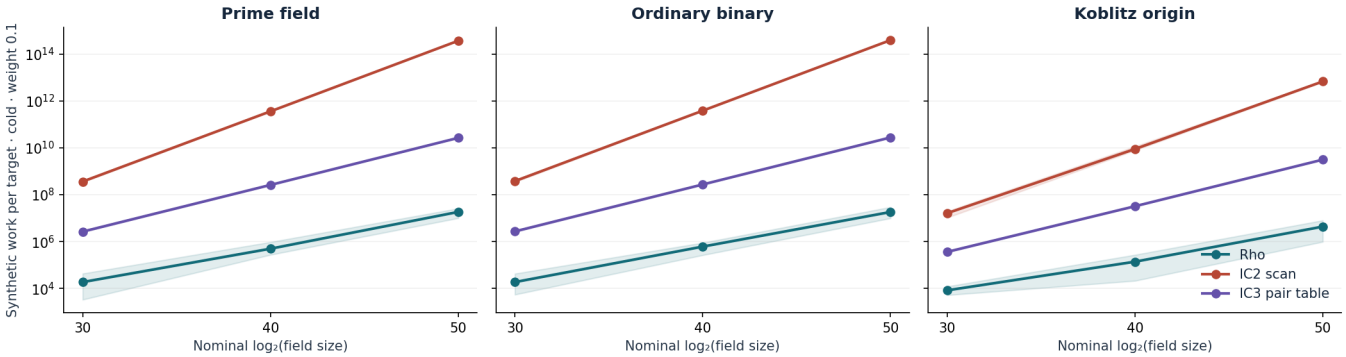
Overlay labels	Multiedges	Mean path length	Diameter
0	160	6.39	8
1	321	4.9	8
2	482	3.74	6
3	643	3.37	6

The degree-3 skeleton has an inert surface and no horizontal degree-3 edges. Other degree labels supply the horizontal connections at every stratum. The cyclic shifts 1, 5 and 7 are illustrative permutations, not calculated actions of actual ideals. Shorter graph paths alone do not change either solver model; the topology-only ablation leaves solver costs fixed by assumption.

A fixed-degree ordinary volcano has horizontal edges only at its surface. The mixed-degree overlay is the explicit correction to the requested picture of horizontal isogenies at all depths.

How size changes the modeled comparison

SYNTHETIC · direct surface cost across size



Seed 20261018 · 10 invented targets per cell · bands = min–max, lines = means · no measured timings

The assumptions determine the scaling

With r the nominal subgroup order, the signed factor-base size is B = 2 ceil(r^(1/3)/2). For m summands, p = 1 − exp(−B^m/(m! r)) is the assumed decomposition probability. Relations needed are ceil(1.08 U) + 8, where U = ceil(B/(2s)) and s is the assumed Frobenius orbit discount.

At fixed ordinary-curve parameters, the scan model makes IC2 collection proportional to r in its leading term. The pair-table model balances a B^2 setup against collection near r^(2/3). Rho uses a square-root collision surrogate. These are imposed model behaviors, not fitted experimental exponents or a comparison with the fastest known implementation.

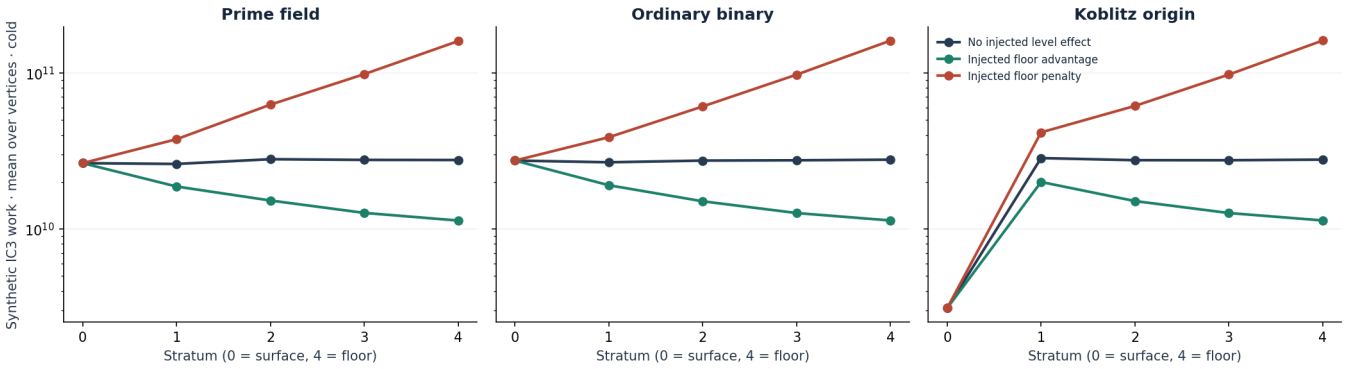
Memory is a separate model output

Size label	IC3 pair-table workspace	Profiled search workspace
30	5,051,720 bytes	10,226,960 bytes
40	507,754,184 bytes	1,015,631,888 bytes
50	51,544,064,000 bytes	103,088,251,520 bytes

Memory is a synthetic capacity estimate; these tables were not allocated. The model charges 24 bytes for every pair-table entry. Profiling keeps the incumbent and current candidate workspaces, plus graph storage. A bounded-memory solver would need a different model.

Does moving to the floor help?

SYNTHETIC · distinguish a null model from a planted trend



50-bit labels · weight 0.1 · per-level vertex counts 1 / 4 / 12 / 36 / 108 · bands summarize 10 target-level means

The null has no injected ordinary-curve level advantage. Its vertex-to-vertex variation is generated from a fixed lognormal multiplier. The green and red trends are deliberate positive and negative controls: they multiply the oracle scan cost by 0.55^level or 1.7^level. Pair-table setup and linear algebra do not receive that multiplier.

A Koblitz origin is not a Koblitz curve at every level

Only the symbolic Koblitz surface receives a Frobenius orbit discount equal to the extension degree. Its descendants use the ordinary-binary model with discount one. A descent can lose the degree-2 Frobenius endomorphism that justified the Koblitz optimization. The graph is a hypothetical depth-4 surrogate: no claim is made that Koblitz curves at exactly extension degrees 30, 40 and 50 realize this graph or its nominal subgroup.

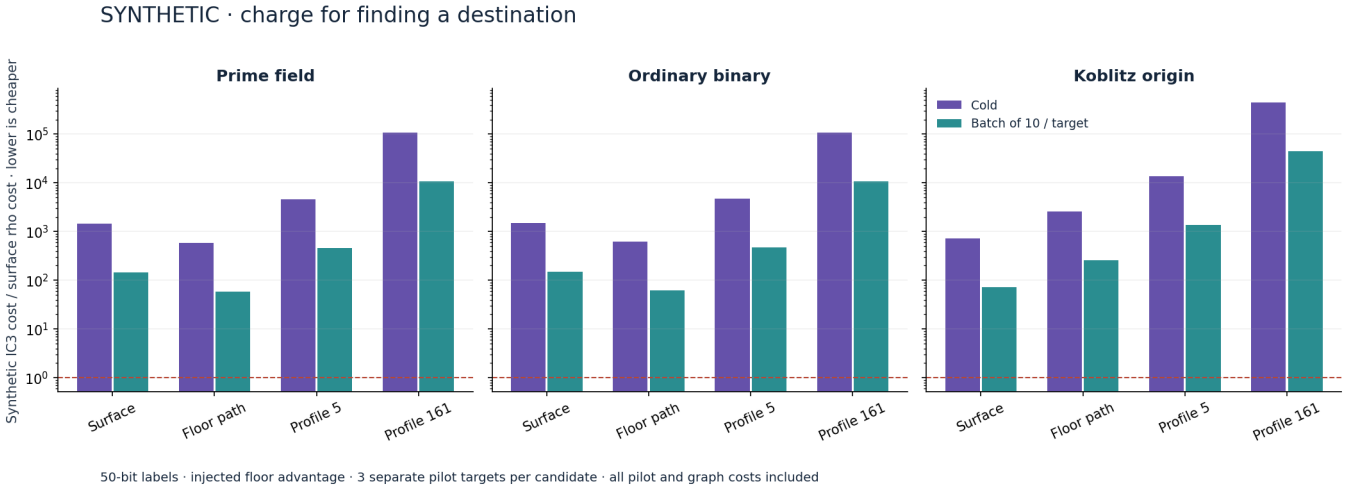
All three requested binary extension degrees are composite. Results from a future real run at these degrees would need separate analysis before any transfer to prime-degree binary extensions.

Avoid a graph-selection artifact

There are 108 floor candidates and only one surface candidate. Under an exchangeable null, 108 of 161 possible minima belong to the floor purely because of candidate count. The package therefore records both profiling all vertices and profiling one prespecified vertex per stratum. Observing a low minimum is not evidence of an intrinsic level effect.

Only ten symbolic targets are requested per cell. The displayed min–max ranges and pattern frequencies are descriptive. They are not confidence intervals for real curves, and the package makes no significance or universal exponent claim.

Count discovery, transport and shared setup



Each bar includes the full modeled pipeline for its accounting mode. Profiling all candidates charges graph construction, all candidate precomputations and all three pilot descents per candidate. The selected candidate is tested on ten separate target labels. Candidate selection never reads those ten test labels.

Prime-field label 50; injected advantage	Cold mean	Batch / target	Cold / rho
Direct surface	2.64e+10	2.63e+09	1.44e+03
Prespecified floor path	1.08e+10	1.09e+09	591
Profile one per level	8.38e+10	8.38e+09	4.57e+03
Profile all 161	1.96e+12	1.96e+11	1.07e+05

The synthetic floor oracle advantage can improve the modeled IC pipeline without making it cheaper than modeled rho. Sharing setup across ten targets changes the accounting; it is reported as batch cost per target and never substituted for a cold solve.

The two-summand and three-summand models use reusable factor-base-log preprocessing with independent target descent. This is a planning model, not a claim that every existing repository IC implementation exposes this reusable interface.

Reproduce and inspect

Build and run from the package directory:

```
bash run.sh run-new
build/simulate --self-test
build/audit run-new
build/simulate --seed 20261018 --out run-replay
diff -qr run-new run-replay
```

The final comparison command applies before reports and audit metadata are added, or should compare the core data files only. Output directories are write-once: the simulator refuses to overwrite an existing run. Fixed-seed replay is byte-identical on the recorded compiler and standard library; C++ distribution algorithms may differ across toolchains.

Validation performed

Check	Status
Structural and accounting assertions	986 passed
Raw observations	130,410 synthetic rows
Independent CSV audit	Checks all raw-to-summary extrema, means, medians and standard deviations; target coverage; pilot/test separation flags
Replay	All core CSV/JSON files compared byte-for-byte on the recorded runtime
Cryptographic verification	Not applicable: no constructed curves, points, maps, relations or recovered logs

trials.csv.gz is the full stage-level observation table. curve_summary.csv.gz retains all per-vertex ten-target summaries. strategy_trials.csv.gz and strategy_summary.csv retain the charged comparisons. level_summary.csv, patterns.csv and pilot_selection.csv retain every control and accounting band. config.json is the frozen model contract.

Requirement status

Requested item	Status within the synthetic scope
30 / 40 / 50 and three families	Complete; all nine label combinations retained
Ten points; min / max / average	Complete as symbolic targets; no actual points or DLP answers
Deep strata and horizontal links	Complete abstract graph; mixed degrees; actual EC realization not attempted
Entire graph comparison, IC focus	Complete for all 161 vertices and three selected solver cost models; all-degree or all-algorithm comparison not claimed
Size-dependent patterns	Complete as outputs of the chosen model; empirical mathematical patterns not established

Repository context and limits

The named repositories supplied the experiment architecture and reporting conventions. Their measured results were not inserted into the synthetic tables, and their cryptographic solvers were not executed. The standard volcano drawing tool is the only retained repository implementation used in producing this package; it is adapted to label abstract vertices and simulated color ranges.

crypto: prime-field IC module

crypto: Koblitz IC module

cryptanalysis: volcano interface

crypto: isogeny/PDP note

crypto: standard renderer

Sutherland: Isogeny volcanoes

Galbraith et al.: subfield index calculus

reference/sources.json pins the inspected file blob hashes. The repository links above may move; the hashes identify the inspected versions.

What a later real experiment must replace

Construct ordinary curves with certified field definitions, orders, subgroup generators and conductor data. Verify every selected isogeny and subgroup transport. Run matched rho and IC implementations on independent targets, recording relation yields, actual rank, oracle work, memory and all setup/transport/verification phases. Replace the assumed cost weights with measured primitive calibrations. Preserve failures and budget limits.

The model intentionally provides no invented coefficients, fabricated isogeny certificates, fake solver correctness flags or synthetic seconds that could be mistaken for runtime measurements. A lower cost in this report demonstrates only the effect of a chosen input assumption.